

Week 7 Analytics Testing & Refinement Report

Data Analytics Track

Project	Improving Patient Appointment Attendance and Healthcare Support Using Data and AI
Dataset	HealthConnect Appointment Data (5,000 appointments)
Primary Tool	Microsoft Power BI
Reporting Stage	Week 7 - Testing, Refinement & End-to-End Validation
Prepared From	Existing HealthConnect analytics dashboard and completed Week 7 testing evidence

Integrity note: A formal Week 6 submission was not completed. This report therefore documents genuine Week 7 testing performed on the existing HealthConnect Power BI analytics component, without claiming uncompleted Week 6 activities.

1. Week 6 to Week 7 Transition

The HealthConnect analytics component available for Week 7 was the existing Power BI dashboard developed from 5,000 appointment records. The dashboard contained an Overview page and an Attendance Drivers page, with Year-Month and Appointment Type slicers and core KPIs for no-shows, cancellations, reminder coverage and repeat no-shows. Because a formal Week 6 submission was not completed, the Week 7 work focused on validating the accuracy and behaviour of this existing analytical component and making a documented usability refinement.

The component selected for testing was the Power BI dashboard and its principal KPI and filter behaviour. The intended purpose of the testing was to confirm that the dashboard continued to present the established baseline values consistently, that the slicers behaved as expected, and that a small refinement did not disrupt dashboard functionality.

2. Testing Objective

The objective of Week 7 testing was to determine whether the main HealthConnect analytical outputs were internally consistent and usable for decision support. Specifically, the tests checked: (1) consistency of the four headline KPI values, (2) correct response of the Year-Month slicer, (3) correct response of the Appointment Type slicer, and (4) continued dashboard functionality after a usability refinement.

3. Testing Approach

- Baseline KPI consistency check: confirm the unfiltered dashboard displays the established KPI values.
- Year-Month filter test: select April, confirm visuals update, clear the filter and confirm baseline values return.
- Appointment Type filter test: select General Consultation, confirm visuals update, clear the filter and confirm baseline values return.
- Dashboard refinement: add a concise “Validated Key Findings” text box to improve interpretation.
- Retest after refinement: repeat the April and General Consultation filter checks to verify that the dashboard remained functional.

4. KPI Validation Results

KPI	Expected / Established Value	Actual Week 7 Value	Result
No-show rate	51.2%	51.2%	Pass
Cancellation rate	5.3%	5.3%	Pass
Reminder coverage	72.7%	72.7%	Pass
Repeat no-show rate	57.8%	57.8%	Pass

All four headline KPI values matched the previously established dashboard baseline. No discrepancy was identified in this consistency check. This test confirms consistency of the displayed KPI values; it does not independently recompute each KPI from raw-row counts.

5. Dashboard Filter Testing

Test	Expected Result	Actual Result	Status	Evidence
Year-Month slicer - April	KPIs and visuals should update; clearing should restore baseline values	Visuals changed when April was selected and original values returned after clearing	Pass	02_Overview_April_Filter.png
Appointment Type - General Consultation	KPIs and visuals should update; clearing should restore baseline values	Visuals changed when General Consultation was selected and original values returned after clearing	Pass	03_Overview_General_Consultation_Filter.png

The two dashboard slicers tested behaved as expected. The visuals responded to each selection, and the dashboard returned to the baseline KPI values after the filters were cleared.

6. Dashboard Refinement

A “Validated Key Findings” text box was added to the Overview page to make the principal validated results easier to interpret. The box summarised the overall no-show rate (51.2%), cancellation rate (5.3%), reminder coverage (72.7%), repeat no-show rate (57.8%), and the fact that the Year-Month and Appointment Type slicers had been tested successfully.

This refinement was intentionally small and evidence-based. It improved the dashboard’s usability by bringing the key validated results together in one clearly labelled location without changing the underlying calculations.

7. Retesting Results

Retest	Expected Result	Actual Result	Status
April Year-Month filter	Dashboard remains responsive after refinement	Filter and visuals operated normally	Pass
General Consultation filter	Dashboard remains responsive after refinement	Filter and visuals operated normally	Pass

Both post-refinement tests passed. The added text box did not interfere with the tested dashboard interactions, providing a documented refine → retest → validate sequence.

8. Validated Findings and Updated Business Interpretation

- The overall no-show rate displayed by the dashboard remained 51.2%, confirming that missed appointments continue to be a major operational issue in the analysed dataset.
- The cancellation rate remained 5.3%, while reminder coverage remained 72.7%.
- The repeat no-show rate remained 57.8%, supporting continued attention to patients with previous no-show history as an important analytical segment.
- The tested slicers functioned correctly, supporting use of the dashboard for segmented review by month and appointment type.

These results support continued use of the dashboard for monitoring and exploratory decision support. However, causal claims should not be made from these tests, and the current Week 7 evidence should be interpreted as dashboard and KPI consistency validation rather than proof that any single factor causes no-show behaviour.

9. Cross-Track Testing Status

Cross-track testing with Data Science was identified as the most relevant dependency because validated analytical findings can support model feature review and interpretation. At the time this report was prepared, no completed cross-track testing result had been obtained. Therefore, this report does not claim completion of the mandatory HC-POD

cross-track validation activity. If a Data Science response is received before submission, the evidence and resulting test/finding/action/retest outcome should be appended to the submission package.

10. Limitations and Remaining Issues

- A formal Week 6 submission and Week 6 advanced-analysis package were not completed, reducing continuity into Week 7.
- The KPI checks in this report validate consistency of the displayed Power BI values against the established baseline; independent raw-data recalculation was not completed as part of the documented tests.
- Cross-track testing with Data Science remained unresolved at the time of report preparation.
- Testing covered the two existing slicers and selected headline KPIs; it was not an exhaustive test of every visual, DAX measure or interaction in the PBIX file.
- The current findings show associations and operational patterns and should not be interpreted as causal effects.

11. Week 8 Recommendations

- Complete the outstanding Data Analytics → Data Science validation and document how the validated analytical findings influence model features, interpretation or decisions.
- Perform an independent raw-data reconciliation for the four headline KPIs where time permits.
- Complete a final visual and interaction review of both dashboard pages before presentation.
- Retain the “Validated Key Findings” summary to support clear communication during final integration and presentation.
- Prepare a concise Week 8 narrative connecting the validated analytics to the wider HealthConnect objective of reducing missed appointments and improving patient support.

Appendix A. Testing Evidence Register

Evidence File	Purpose
01_Overview_No_Filter.png	Baseline dashboard evidence
02_Overview_April_Filter.png	Year-Month slicer test evidence
03_Overview_General_Consultation_Filter.png	Appointment Type slicer test evidence
04_Dashboard_Refinement.png	Evidence of the Validated Key Findings refinement
05_April_Retest.png	Post-refinement April retest evidence
06_General_Consultation_Retest.png	Post-refinement General Consultation retest evidence
HealthConnect_Week7_Testing_Record.docx	Structured testing and validation record